

MAT 1272 – Group Activity

Chapter 1 Review: Variables, Population, and Sample

Work with your group. Discuss your answers and be ready to explain your reasoning.

1. What are we studying?

A professor collects the following information from students: *major, number of siblings, hours spent studying each week, favorite mode of transportation, height, number of courses taken.*

1. What is a variable?
2. For each item, decide whether it is qualitative/categorical or quantitative.
3. For each quantitative variable, decide whether it is discrete or continuous.
4. Choose one variable your group thinks is tricky. Be ready to explain your reasoning.

2. Classify the variables

For each variable, write **Q** (quantitative) or **C** (qualitative/categorical). If quantitative, write **D** (discrete) or **Co** (continuous).

Variable	Q/C	D/Co
1. Number of students in a class		
2. Eye color		
3. Amount of money in a bank account		
4. Number of text messages sent yesterday		
5. Time needed to complete an exam		
6. Type of phone owned		
7. Weight of a student		
8. Number of cars in a parking lot		

Group discussion: What is the difference between a discrete variable and a continuous variable?

3. Population and Sample

A professor wants to know how many hours per week **City Tech students** spend studying mathematics. She asks **40 MAT 1272 students** how many hours they study each week.

1. What is the population?
2. What is the sample?
3. What is the variable?
4. Is the variable qualitative or quantitative?
5. Is the variable discrete or continuous?

4. Think like a statistician

For each situation, identify the **population, sample, and variable**, and classify the variable.

A. A researcher wants to know the average amount students spend on lunch at City Tech. She asks 75 students how much they spent on lunch that day.

B. A professor wants to know how many hours students spend studying. She asks 30 students from her class.

C. A college wants to know students' preferred transportation to campus. It asks 100 students whether they walk, take the subway, take the bus, drive, or use another method.

5. Challenge your group

Create **two variables**: (1) one qualitative/categorical variable; (2) one quantitative variable. Decide whether your quantitative variable is discrete or continuous. Be ready to explain your choices.

Before you finish: Can everyone in your group explain variable, qualitative, quantitative, discrete, continuous, population, and sample?